

Tooling Purchase Order Accrual Processing

Finance • Automated Accrual System

Version 1.0

1. Purpose

This SOP defines the automated process for calculating and recording tooling accruals for open purchase orders. The system replaces the manual 8–11 day monthly Excel process with intelligent SAP integration and materiality-based decision making.

2. Business Context

Problem Statement

Manual tooling accrual calculations are taking 8–11 days monthly, causing:

- Human calculation errors
- Delayed financial close processes
- Earnings volatility from late accruals

Solution

Automated materiality-based processing with intelligent decision hierarchy:

- **\$150K threshold** — primary decision point for workflow routing
- **\$300K threshold** — high-value processing with extended inquiry periods
- **WBS integration** — project milestone dates from business finance
- **Straight-line amortization** — monthly accrual calculations

3. Mandatory Rules

Rule	Requirement
No Data Fabrication	Never make up or simulate SAP data. All document numbers, dates, and transaction data must come from actual SAP.
Accrual Calculation	Always: $\text{Monthly Rate} \times \text{Time Elapsed}$. Never use full Outstanding Balance. If accrual = outstanding balance and time = 0, then accrual = 0.
Journal Entry Posting	No workflow auto-posts. All workflows require finance team approval. Only after approval should entries be parked in SAP.
SAP API Failure	Document exact error. Update case to error. Notify finance. Do NOT proceed with fabricated data.
Approval Required	Submit approval request ticket with clear summary. Stop processing. Reviewer decision triggers next step automatically.

4. Decision Hierarchy

Condition	Workflow	Action
PO Amount > \$300,000	EMAIL_INQUIRY	Contact PO owner for completion date confirmation (~6 mo response)
PO Amount ≥ \$150,000 AND has WBS Element	PROJECT_MILESTONE	Use project milestone dates from business finance
PO Amount ≥ \$150,000 AND no WBS Element	EMAIL_INQUIRY	Contact PO owner for completion date confirmation
PO Amount < \$150,000	DELIVERY_DATE	Use SAP "Need by" date from scheduling lines

5. Mandatory First Steps — All Workflows

Step 1 — Retrieve Actual SAP Data

Before any processing, retrieve ALL required data from SAP APIs:

- **PO Header:** PO Number, Total Amount, Creation Date, Creator Email, Company Code
- **Line Items:** Line #, Order Qty, Unit Price, GL Account, Cost Center, Description
- **Schedule Lines:** Delivery Date, Schedule Line #, Confirmed Qty
- **Account Assignment:** WBS Element (if exists), Cost Center, GL Account
- **Invoice History:** All posted supplier invoices — numbers, amounts, posting dates

Step 2 — Calculate Outstanding Balance

Using the actual SAP data:

- Total PO Value = Order Quantity × Unit Price
- Total Invoiced Amount = SUM of all invoice amounts
- **Outstanding Balance = Total PO Value – Total Invoiced Amount**

Decision Points:

- If Outstanding Balance = \$0 → **STOP** (fully invoiced)
- If Outstanding Balance < \$0 → **FLAG** for manual review (over-invoiced)
- If Outstanding Balance > \$0 → **CONTINUE** with workflow decision tree

Step 3 — Determine Workflow Path

Use ACTUAL Outstanding Balance (not total PO value) for all threshold comparisons. Check for WBS Element presence and follow the Decision Hierarchy above.

6. Workflow Implementations

Workflow 1 — EMAIL_INQUIRY

Trigger: PO > \$300K OR (PO ≥ \$150K AND no WBS) • Avg response: 6 months

1. Send inquiry email to PO owner asking for completion date (YYYY-MM-DD)
2. Wait for response — update case state
3. Parse & validate completion date from response
4. Update PO delivery date in SAP schedule lines
5. Calculate accrual using new delivery date
6. Validate calculation, send to Finance for approval
7. Once approved → park journal entry in SAP
8. Send confirmation with document number

Workflow 2 — PROJECT_MILESTONE

Trigger: PO ≥ \$150K AND has WBS • Typical lifecycle: 12 months

1. Extract WBS element from PO account assignment
2. Query project system for milestone dates (Project → WBS → Project end)
3. Fallback: use PO delivery date if WBS data unavailable
4. Calculate accrual, validate, send to Finance for approval
5. Once approved → park journal entry in SAP

Workflow 3 — DELIVERY_DATE

Trigger: PO < \$150K • Duration: based on actual SAP delivery date

1. Extract delivery date from SAP schedule lines
2. Validate date (not in past, not > 5 years future)
3. Calculate accrual, validate, send to Finance for approval
4. Once approved → park journal entry in SAP

7. Accrual Calculation Process

Step	Formula / Action
1. Determine PO Balance	Outstanding Balance = Total PO Value – Invoiced Amount
2. Determine Time Period	Start: PO creation date • End: from workflow Total Duration in months
3. Monthly Accrual Rate	Monthly Rate = Outstanding Balance ÷ Total Duration (months)
4. Current Month Accrual	Accrual Amount = Monthly Rate × Time Elapsed Time Elapsed = (Month End – PO Creation) in months

Example Calculation

Parameter	Value
PO Creation Date	November 24, 2025

Parameter	Value
Delivery Date	February 24, 2027
Month End Date	November 30, 2025
Total PO Value	\$500,000
Invoiced Amount	\$0
Outstanding Balance	\$500,000
Total Duration	15 months
Monthly Rate	\$33,333.33
Time Elapsed	0.2 months (6 days)
Accrual Amount	\$6,667

8. Accrual Calculation Validation

Before sending ANY email to Finance, validate:

1. Month-end date is calculated dynamically (not hardcoded)
2. Time elapsed is in months (including fractional months)
3. Monthly Rate = Outstanding Balance ÷ Total Duration
4. Accrual Amount = Monthly Rate × Time Elapsed
5. Accrual Amount < Outstanding Balance (unless time = total duration)
6. Accrual percentage makes sense (e.g., 6 days / 14 months ≈ 1.4%)

■ Common Mistake	✓ Correct Approach
Using Outstanding Balance (\$450K) as Accrual Amount → 100% of balance!	Monthly Rate × Time Elapsed = \$32,143 × 0.2 = \$6,429 (≈1.4%)

9. Journal Entry Parking

After receiving finance approval, park the journal entry in SAP using **ZACC_DOC_SRV** API with the calculated accrual amount and PO number as reference. This creates a parked document (Dr. 1011000 Tooling Assets / Cr. 2010131 Accrued Expenses) ready for final posting.

10. Error Handling & Communication

For any SAP errors, include in all communications:

- Error Type & Message
- SAP Error Code
- Timestamp, PO Number, Case ID
- Workflow step, SAP API endpoint

- Impact description & required next steps

11. Contacts

Role	Contact
PO Owner	{{CONTACT_PO_OWNER}}
Finance Team	{{CONTACT_FINANCE_TEAM}}
Technical Support	{{CONTACT_TECHNICAL_SUPPORT}}